

Dark matter and dark energy as field states: The Universal Quantum Foam Hypothesis as an ontological framework

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Abstract

The Universal Quantum Foam Hypothesis (UQSH) denotes a higher-level ontological framework within which the Quantum Foam Hypothesis (QSH) is used as a concrete field-mechanical description of physical space. In this view, space is not a passive background, but a continuous, dynamic field medium ("Qu-foam") whose local and large-scale state is characterized by matter, radiation, and energy. Gravity is not primarily interpreted as a function of an additional, invisible material component, but rather as the geometric and dynamic response of the field medium to stable or metastable states.

The UQSH thus does not claim to replace established field equations, but instead integrates known phenomena into a unified field-mechanical interpretation: Dark matter effects appear as expressions of fieldborne prestresses and persistent stress gradients, while dark energy is understood as the relaxation tendency of the field. The aim is an interpretation that does without new fundamental particles, respects the local invariance of c , and at the same time reads observational findings in such a way that testability and falsifiable consequences are paramount.

1 Radiation-induced field bias, dark matter effects and the saturation limit of a dynamic space field

The starting point of the Universal Quantum Foam Hypothesis is the insight that physical space is not empty, but consists of a finely structured, dynamic medium that behaves like a foam of microscopic field zones. These "foam bubbles" are not particles or discretely defined building blocks, but local states of organization within the field, in which motion, tension, and relaxation are bound and transmitted. They form the elementary texture of space. Space is thus not a mosaic of solid cells, but a continuous network of interlocking field states that can shift against one another, superimpose, merge, and unravel. Every radiation, every movement, every material bond does not act in an abstract vacuum, but in this quantum foam, deforming it, pre-tensioning it, and leaving traces in its geometry and thus in its gravitational effect. In this sense, space is not a passively given container, but an active, reactive medium capable of carrying and preserving states.

These field distortions do not exist solely in two clear states of "free" and "bound." Between these two lies a broad spectrum of metastable configurations. Some distortions dissipate immediately, while others persist for extended periods without yet appearing as matter in the strict sense. The Qu foam is therefore permeated by short-lived, gliding, oscillating, and semi-bound patterns that pre-structure the field and influence its response to new excitations. Radiation oscillations propagate isotropically and spherically through this medium from the point of emission. They are not directed arrows, but rather spherical field responses. When such a wavefront encounters a local field node or a quantum-like focal point, this also occurs isotropically. The incoming interferences do not superimpose at a single point, but instead generate minute spherical distortions within the medium. Each reception is itself a local emission. The field responds to every excitation with a new, small-scale sphere of tension and relaxation. This creates a dense network of interlocking, overlapping, and partially persistent deformations.

In this way, large-scale landscapes of tension build up. A star, a galaxy, or an active nucleus generates local pre-stresses in the medium, which spread, flatten out, and intertwine with other pre-stresses. Where many such influences converge, overarching field structures emerge. Space does not become "heavier" in the material sense, but rather geometrically more densely organized, stiffer, and more strongly structured. The superposition of these deformations is not additive, as with numbers, but geometric. Tensions can intensify, redirect, subside, or transform into new patterns. No sharp boundaries arise between neighboring field states, but rather zones of shared organization. From this perspective, galactic halos are not mere shells of matter, but extended superposition spaces of such field deformations, continuous tension profiles of the medium itself, fed by the sum of countless past and present stimuli.

On November 8, 2025, it became clear for the first time that preci-

sely this field-mechanical pre-stress of space could explain the observed dark matter effects. The central idea was that gravitational effects need not be bound to an invisible substance, but could be understood as persistent states of stress within this Qu-foam. At that moment, radiation no longer appeared merely as transported energy, but as real field excitation that structures space itself. Galaxies, stars, and active nuclei shape the Qu-foam not only locally, but across very large scales. Space carries these imprints onward, even when their immediate sources are no longer active. In this view, the so-called halo is not an object, but the geometric perpetuation of field-mechanical processes.

To move this interpretation beyond a purely qualitative level, it is useful to explicitly link the gravitational potential within the UQSH framework to the underlying field state.

2 Effective Gravitational Potential within the UQSH Framework

Within the Universal Quantum Foam Hypothesis (UQSH), gravitation is not understood as an independent fundamental interaction, but as the dynamical response of a continuous field medium to local inhomogeneities of its tension state.

Physical space is characterized by a spatially dependent field state $Qu(x)$ which describes the local tension or excitation level of the medium. Time dependence $Qu(x, t)$ is in principle possible and relevant for dynamical processes; however, for the minimal formalization of the effective gravitational potential introduced here, the quasistatic representation $Qu(x)$ is sufficient.

Within this framework, the classical gravitational potential $\Phi(x)$ can

be directly identified with the field tension state. We set

$$\Phi(x) = \alpha Qu(x), \quad (1)$$

where α is a dimension-dependent proportionality factor (not further specified here).

The local reorganization response of the field to tension inhomogeneities gives rise to the observed acceleration:

$$\vec{a}(x) = -\nabla\Phi(x) = -\alpha \nabla Qu(x). \quad (2)$$

In the weak-field regime, this identification is compatible with the classical Poisson formulation; instead of a substantial mass density $\rho(x)$, the source term is given by the density of bound field organization:

$$\nabla^2 Qu(x) \propto \rho_{\text{bind}}(x), \quad (3)$$

where $\rho_{\text{bind}}(x)$ is not to be interpreted as a substantial mass density, but as the locally measured degree of stabilized field binding.

This perspective allows gravitational excess effects to be interpreted as consequences of the total distribution of bound field organization. Deviations between visible matter and effective gravitational influence do not necessarily arise from missing matter, but from additional contributions to the field tension that need not correlate directly with baryonic luminosity.

3 Relation to Dark Matter Phenomenology

Deviations between observed gravitational effects and luminous matter distributions can arise directly from non-luminous contributions of bound field organization. Phenomena commonly attributed to dark matter may

thus be interpreted as manifestations of nonlocalized field binding, without the introduction of additional particle species.

The following sections discuss observational implications of this identification (rotation curves, gravitational lensing, clusters), with the above formalization providing the basis for the analysis.

A second, necessary step arose directly from this insight. If space is a physical medium, then its deformation cannot increase indefinitely. A real field possesses elasticity and therefore also limits. As early as November 9, 2025, it became clear that the Qu-foam has a maximum stress absorption capacity. Once this limit is reached, the medium no longer reacts with further local compression, but rather with spatial expansion and redistribution. This saturation limit explains why dark matter effects do not increase arbitrarily, why galactic centers remain flat, why halos are limited, and why extreme compressions do not lead to mathematical singularities.

The radiation-induced DM effect and the field-mechanical saturation limit are thus inextricably linked. The former describes how the field, through countless isotropic emissions, superpositions, and metastable deformations over large scales, reaches a prestressed state. The latter explains why this prestress assumes stable, finite forms. The Qu foam is neither a passive background nor an arbitrarily malleable continuum. It is an active medium with internal structure and physical limits. In this context, gravity does not appear as the effect of hidden matter, but as the dynamics of a space that itself carries tension, is organized, and reacts to stress. Motion does not follow a force acting at a distance, but rather the geometry of this prestressed medium. A body falls because it follows a tension gradient in the field. Gravitation is the form that space assumes when it carries structure, and the expression of the fact that space itself is the true dynamic subject of physics.

4 Observational findings and their field-mechanical interpretation

Key observations in modern astrophysics show that gravitational effects in galaxies and galaxy clusters appear systematically decoupled from baryonic mass. Flat rotation curves, the close coupling between baryonic distribution and dynamic acceleration, precise scaling relations such as the baryonic Tully–Fisher relation, strong gravitational lensing beyond the visible mass, and the separation of gas and lensing centers in systems like the Bullet Cluster form a consistent pattern of gravitational anomalies. In the standard framework, these findings are interpreted as evidence for an additional, non-collision matter component. Within the UQSH, however, the same data appear as a natural consequence of a dynamic space field: Additional gravitational effects do not primarily follow mass, but rather the stability, coherence, and superposition of field-mechanical patterns. Baryonic structures act as anchors that shape the local stress state of the field; large-scale superpositions of these stress states generate persistent geometric profiles that manifest as rotational dynamics and lensing effects. The fact that the "DM signature" couples so closely to visible structures is, in this view, not a fine-tuning but rather an expression of a shared field-mechanical basis. The center of gravity follows the most stable patterns, not necessarily the largest baryonic mass, which is why turbulent gas can lose its gravitational dominance in collisions, while coherent galactic structures retain it. The observed anomalies remain completely intact; what changes is their ontological interpretation: not as the effect of an additional substance, but as the geometry of a stressed medium.

5 Radial Acceleration Relation

The radial acceleration relation describes one of the most precise empirical correlations in modern galaxy physics. It shows that the actually observed

dynamic acceleration in galaxies follows almost exactly the acceleration calculated from the baryonic mass distribution. In other words, where visible matter is highly concentrated, the so-called "dark matter effect" is also strong; where the baryonic structure is diffuse, the deviation is small. This relation holds true over many orders of magnitude and for very different types of galaxies, from dwarf galaxies to massive spirals, with remarkably little variation.

Within the CDM framework, this behavior is not self-evident. If dark matter is an independent, collisionless substance distributed in large-scale halos, then its gravitational effect should only correlate indirectly with the baryonic structure. In practice, this close coupling must be explained by complex origin histories, feedback processes, and statistical fine-tuning. The RAR appears there as an emergent byproduct of galaxy evolution, not as a fundamental law. It is precisely this need for "readjustment" that makes the relationship in need of explanation.

Within the framework of UQSH, the RAR is not a surprise, but rather the expected normal state. The so-called halo is not a separate substance here, but a stress gradient of the Qu foam, i.e., a geometric response of the space field to stable bound patterns. Baryonic structures are themselves field-mechanical states of high coherence. They directly deform the local and large-scale state of the medium. In this view, the "dark matter effect" is nothing other than the field-wide reaction to these bound patterns.

Thus, the additional gravitational effect inevitably follows the visible structure. Compact baryonic configurations generate steep stress profiles in the field, while diffuse or ordered structures lead to flatter and smoother profiles. The field geometry reflects the geometry of matter because both are expressions of the same medium. There is no second, independent component that would need to be subsequently adjusted. The close coupling between baryonic distribution and dynamic acceleration

is therefore not a fine-tuning process, but a direct consequence of their shared field-mechanical basis.

From this perspective, the RAR transforms from an anomaly into a key finding. It shows not that two fundamentally separate components cooperate remarkably well, but that gravity and matter emerge from the same physical substrate. The relation suggests that what appears as "dark matter" is not a hidden substance, but rather the continuous geometric response of a dynamic spatial field to stable material patterns.

6 Baryonic Tully–Fisher-relation

The baryonic Tully-Fisher relation describes one of the most stable empirical laws in galaxy physics: the asymptotic rotational speed of a galaxy scales with remarkable precision with its total baryonic mass. Over many orders of magnitude, galaxies with more visible matter systematically rotate faster, with a dispersion that is significantly smaller than predicted by simple formation models. This relation connects two quantities that, in the standard picture, are of fundamentally different origins: baryonic matter on the one hand, and gravitational dynamics, which in the CDM framework is largely determined by a separate dark matter component, on the other.

Within standard cosmology, the baryonic mass distribution (BTFR) is not understood as a direct consequence of postulated dark matter. It arises statistically across populations and is interpreted as a result of the co-evolution of baryonic and dark matter, mediated by accretion history, feedback processes, star formation, and environmental dependencies. The high precision of this relationship is therefore often considered surprising, as it implies a close alignment between two ontologically separate components. Within this framework, the reason why a fundamentally

independent dark matter distribution so accurately "mirrors" the baryonic mass is not a direct consequence of the basic assumptions, but rather an emergent result of complex evolutionary processes.

Within the framework of UQSH, this surprise loses its character. The baryonic Tully–Fisher relation appears here as a direct consequence of the field-mechanical coupling between matter and gravity. In this model, baryonic mass is not merely a collection of particles, but a stable, coherent field state that structures the local and large-scale stress state of the Qu foam. Every baryonic structure is simultaneously a geometric modulation of the space field.

With increasing baryonic mass, these field modulations overlap more strongly. Many local stress contributions result in a deeper, continuous stress gradient in the medium. From this perspective, the rotational speed of a galaxy does not reflect the presence of a separate, invisible substance, but rather the field's response to the superposition of bound patterns. The more baryonic structure is present, the more pronounced the field-mechanical prestress, and the higher the characteristic rotational speed.

The BTFR thus becomes a direct geometric relationship: more bound field structure leads to stronger stress superposition, which in turn creates a deeper field-wide "trench," and this manifests dynamically as a higher rotational speed. The relationship is not a fine-tuning of two independent components, but rather an expression of the same physical basis in two different interpretations: once as material structure, and once as a dynamic field response.

From this perspective, the BTFR does not measure the interplay of visible matter and hidden substance, but rather the unity of matter and space field. It shows that gravity in galaxies is not an addition to baryonic

structure, but its immediate geometric unfolding within the medium. The close coupling is therefore not a mystery, but an indication that both quantities emerge from the same ontological basis.

7 Gravitational lensing

Gravitational lensing measurements are among the most direct approaches to the largescale geometry of space. They reveal how strongly light rays are deflected as they travel through galaxies or galaxy clusters. In many systems, this deflection is significantly greater than would be expected from the observable baryonic mass. Within standard cosmology, this finding is interpreted as direct evidence for an additional gravitational source, which is postulated to be in the form of extended dark matter halos. Since the lensing strength is proportional to the gravitational effect integrated along the line of sight, the discrepancy between the lensing effect and the visible mass is considered one of the most robust empirical supports for the dark matter hypothesis. This effect is particularly pronounced in galaxy clusters and galaxies, where the observed deflection systematically increases with the number and spatial density of the galaxies involved.

Within the framework of UQSH, this scaling is interpreted fundamentally differently. Light is not understood here as a stream of particles reacting to isolated masses, but rather as the boundary dynamics of the field itself, propagating along the energetically favored geometry. It "reads" the structure of the medium. In this view, the deflection of light is not a reaction to additional substance, but to the geometric organization of the spatial field.

Each galaxy represents a stable, bound field state that alters the local stress state of the Qu foam. These changes are not point-like, but spatially extended. In galaxy clusters, the stress profiles of many such field states overlap. This creates a coherent, deepened geometric stress

field that does not correspond to the sum of individual mass points, but to the collective organization of the medium. The field does not "sense" how much baryonic matter is contained in a particular object, but rather how strongly and coherently space as a whole is stressed.

In this model, light propagates along the geodesics of a physically stressed medium. It therefore does not primarily react to individual galaxies, but rather to the global field geometry created by their collective presence. As the number and density of galactic structures increase, this geometry deepens continuously. The "trench" through which the light travels becomes deeper, not because additional matter is present, but because the field tension is superimposed over a large area. The lensing effect thus increases systematically, without the need to introduce any further material component.

From this perspective, strong gravitational lensing is not an indication of invisible mass, but rather an expression of the persistence and superposition of field-mechanical stress states. Galaxy clusters act as optically powerful lenses because they represent collective nodes of the space field in which many stable patterns geometrically cluster. The observed scaling of the lensing effect with the complexity and density of a system thus becomes a natural consequence of a dynamic medium that carries, accumulates, and coherently organizes stresses over large distances.

In this view, gravitational lenses do not measure hidden matter, but rather the actual shape of space. They show that gravity is not limited to objects, but resides within the medium itself. What appears as an "excessive" deflection is not an excess of mass, but rather an indication that the field can support more than the visible matter alone would suggest.

8 Bullet Cluster

Observations of the so-called Bullet Cluster are considered one of the strongest arguments for the existence of non-collision dark matter in standard cosmology. When two galaxy clusters collide, the hot, baryonic gas remains behind in the collision zone because it interacts strongly and is slowed down, while the galaxies and stars continue largely unimpeded. Gravitational lensing measurements show that the centers of gravitational influence do not coincide with the gas left behind, but rather follow the galaxy populations. Within the CDM framework, this leads to the conclusion that the majority of the gravitationally effective mass is not bound to the collision-prone gas, but to a non-collision component that travels through the system along with the galaxies.

Within the framework of UQSH, this finding takes on a different meaning. Here, gravity does not primarily follow the amount of baryonic matter, but rather the stability and coherence of field patterns. Material structures do not act solely through their mass, but by representing stable, bound states of the Qu foam. During such a collision, the hot gas is in a highly turbulent and unstable state. Shock fronts, mixing, and dissipation largely destroy its coherent structure. It thus loses a significant portion of its ability to maintain a consistent state of tension in the field. Its field-mechanical imprint becomes flat, fragmented, and localized.

Stars and galaxies, on the other hand, remain comparatively stable patterns even under extreme dynamic conditions. They retain their internal structure, their orbital dynamics, and their coherent coupling to the field. As long-lived, compact states of the Qu foam, they carry on the associated state of stress. In the UQSH, it is precisely these stable patterns that act as field-mechanical anchors. Space does not organize itself around the greatest instantaneous mass concentration, but rather around those structures that preserve their geometric identity over time.

From this perspective, the observed center of gravity does not follow the remaining gas, but rather those patterns that maintain their coherence through the collision. The separation of gas and lens center in the Bullet Cluster is thus not necessarily evidence of an invisible substance, but rather an expression of different field-mechanical responses of stable and unstable structures in a highly dynamic process. Turbulent, dissipative matter loses its geometric influence, while long-lived, compact patterns retain it.

What appears in the standard picture as "proof of collisionless matter" becomes, in the UQSH, a demonstration that gravitational action is coupled to the stability of field states. Gravity does not simply follow mass, but rather the ability of a structure to maintain a coherent state of tension in the space field. The Bullet Cluster thus does not primarily show that there must be an invisible substance, but rather that space itself distinguishes between stable and unstable patterns.

9 Saturation, substructure, and the cosmic web as field geometry

Several open problems in standard cosmology, such as the core-cusp problem, the too-big-to-fail problem, missing satellites, and the limited depth of gravitational halo profiles, indicate that gravitational extraneous effects do not increase indefinitely. Simulations often predict steep, cuspy centers, while observations in many galaxies show flat cores. Likewise, gravitational substructures exist without visible galaxies, and conversely, many of the expected satellites are missing. In the CDM framework, these discrepancies are addressed by complex baryonic feedbacks. UQSH, on the other hand, interprets them as a direct consequence of the finite elasticity of the space field.

10 Core–Cusp-Problem

The core-cusp problem describes the discrepancy between numerical simulations of cold dark matter, which typically predict steeply rising, peaked density profiles in the centers of galaxies, and observations, which in many cases show significantly flatter cores. Within the CDM framework, this discrepancy is usually explained by baryonic feedback processes, such as repeated gas outflows, star formation, or supernova-induced feedback, which are thought to subsequently smooth the initially steep profiles. These approaches are possible in principle, but remain highly model-dependent and require finely tuned assumptions about the efficiency, time histories, and coupling strengths of the processes involved.

Within the framework of UQSH, the flattening of the central profiles results from the fieldmechanical properties of the medium itself. If halos are understood not as material density distributions but as stress gradients of the Qu foam, an infinitely steep central region is not natural. The field possesses finite elasticity. Above a local stress threshold, additional loading no longer leads to further deepening but to a spatial redistribution of the field stress. The profiles flatten in the center and taper gently outwards. Flat cores thus appear not as an exception but as a field-mechanically preferred state.

In this context, gamma radiation represents the extreme case of field relaxation. It arises where a highly saturated, strongly bound field state abruptly abandons its geometric identity. In the language of UQSH, it is the echo of maximum field tension that can no longer be maintained and reverts to nearly free limit dynamics.

For the halo and core-cusp problems, this means that gamma events are local discharges of field overload. They indicate that the medium has reached a limit of its elasticity. In a purely matter-based view, such

events appear as high-energy particle processes. From a fieldmechanical perspective, they are manifestations of the fact that the Qu foam is forced to release excess stress in certain regions.

This property prevents the formation of arbitrarily steep centers. Where the field tension becomes too great, it cannot be further accumulated"but instead tips over into discharge. Gamma radiation is not the cause but a symptom of this transition. It is the form in which the field releases its overload. Thus, it indirectly acts as a stabilization mechanism: it limits the local field density and helps prevent the formation of infinitely large cusps.

One could say: matter binds field tension. Gravitational halos are large-scale tension profiles. Gamma radiation is the moment when the field can no longer support a local supersaturation and returns it to free motion.

In this sense, gamma radiation is not a special phenomenon of extreme astrophysics,"but rather the visible limiting process of field elasticity. It indicates that the medium itself possesses an inherent dynamic limiting mechanism. The same dynamic that appears as highenergy radiation on small scales acts as a natural flattening of stress profiles on large scales. Thus, gamma radiation connects the micro level of extreme processes with the macrostructure of galactic fields, revealing that the medium cannot be compressed indefinitely.

11 Too-Big-To-Fail and Missing Satellites

In this view, gravitational substructure appears not as a collection of invisible objects, but as continuous zones of tension within a coherent field. Subhalos can be mechanically "massive"without baryonic condensation. Visible satellite galaxies only form where additional stability conditions are met. Too-Big-To-Fail and Missing Satellites thus become two aspects

of the same principle: gravitational effect and material manifestation are not identical.

On a large scale, this field geometry becomes visible in the cosmic web. Filaments, knots, and voids appear in the UQSH not as a representation of an invisible matter distribution, but as favored stress paths of the space field itself. They act like elastic guiding structures along which matter organizes itself without being material substance itself. Galaxies follow these paths because they represent energetically favored configurations in the field. Large-scale magnetic structures can be integrated into this geometry, not as the primary cause, but as mesoscale manifestations of the same field-mechanical order.

This fundamentally shifts the perspective on cosmic structure. The universe no longer appears as a stage upon which visible and invisible matter are distributed, but as an active medium that itself forms geometric order. Gravitational effects are not signatures of hidden entities, but rather expressions of the organization, saturation, and persistence of a dynamic spatial field.

12 Dark energy and redshift

In the UQSH, dark energy does not appear as an independent substance, nor as a mysterious additional force, but rather as a field-mechanical relaxation tendency of the Qu-foam itself. It does not denote an actor, but a regime state of the medium. In low-tension, structurally weak areas, the fundamental tension of the field can unfold more freely. Dark energy is thus not "something" that acts, but rather a description of what an elastic continuum does when it is not locally bound: it relaxes on a large scale. While the so-called dark matter effects arise from local and regional tension states of the field, dark energy describes the complementary

boundary process, the large-scale tendency of the medium to rebalance overstretched or asymmetrically loaded states. Where the field binds, it must release elsewhere. Order and relaxation are two sides of the same dynamic.

This relaxation is not a localized event, but rather a planar reorganization of the field geometry. It does not accelerate objects through pressure, but rather alters the metric organization of space itself by shifting the field structure towards lower overall tension. This is particularly evident in voids. There, the coupling and disturbance density of the field is lower, the local prestress is weak, and the field elasticity is high. In UQSH, voids are not "empty spaces," but rather zones of increased relaxation freedom. In them, the fieldmechanical dynamics run faster, processes relax more quickly, and time appears accelerated in its internal rhythm. In this sense, dark energy is not a cosmic engine, but the natural mode of motion of a medium that is not uniformly bound everywhere.

This relaxation must be strictly distinguished from the observed redshift. It arises from the propagation of radiation patterns in the Qu foam, a process that is radio-mechanical. Radiation is not a point-like entity from the outset, but rather a spatially and spherically distributed excitation structure. With increasing distance from the emission point, the same excitation is distributed over an increasing volume of the field. The excitation density per volume element decreases. The observed frequency in the UQSH is directly coupled to this field-mechanical density. If the excitation is diluted, the effective wavelength lengthens. Redshift is therefore an expression of a decreasing excitation density in the medium, not the result of an external cosmic process.

This basic effect, however, is not homogeneous. Along long lines of sight, radiation passes through different field-mechanical regimes. In structured regions, the field is more densely coupled, and excitations are more

frequently rearranged and perturbed. In voids, on the other hand, the coupling density is lower. Radiation patterns are less restructured there and appear more effectively longer-wavelength. Passage through voids therefore increases the redshift section by section, without dark energy itself causing the frequency shift. Over cosmological distances, a sequential superposition occurs: normal field propagation in structured regions, increased rarefaction in void zones, normal propagation again, the next void passage, and so on. The radiation arriving at the observer carries a cumulative, pathdependent, higher-order redshift resulting from the repeated passage through differently strained field regions.

From this perspective, blueshift and redshift appear as two sides of the same fieldmechanical principle. Blueshift occurs where radiation excitations become denser, for example, near compact or highly excited field states. The increased excitation density leads to shorter effective wavelengths. Redshift arises where radiation patterns become more dilute than would be expected from purely geometric propagation, for example, due to large-scale field relaxation or repeated void passages. Frequency changes are therefore not independent, separate processes, but direct manifestations of the field-mechanical density of the excitation.

Dark energy and redshift belong to the same ontological context within this framework, without being identical. Dark energy describes the large-scale relaxation state of the medium itself. Redshift describes how radiation excitations are spatially distributed and fieldmechanically diluted within this medium. Discrepancies between local and large-scale cosmological distance measurements can be understood in this view as an indication that cosmological lines of sight are influenced to varying degrees by voids and structure-rich regions, and that path-dependent contributions to redshift are physically relevant. The observed effects do not arise from an external dynamic or an additional driving entity, but from the internal state structure of the field itself. Space is not uniformly

bound everywhere. It carries tension, it possesses relaxation regimes, and it reacts to these differences with measurable effects on radiation and dynamics. In this sense, dark energy is not a driving mechanism, but rather the term for a field-mechanical state in which relaxation predominates over binding.

13 Ontological shift and scientific consequences

The key difference between standard cosmology and the UQSH lies not in the observed phenomena themselves, but in their ontological interpretation. Rotation curves, gravitational lensing, substructure in halos, the cosmic web, the tight couplings between baryonic matter and dynamic acceleration, and the observed redshift over cosmological distances all remain completely intact. What changes is the assumption about what causes these effects. While the standard model interprets them as evidence of additional, as yet undetected substances or independent cosmic actors, the UQSH understands the same findings as properties of an active space field that can generate, sustain, confine, and release tension on a large scale.

Gravitational action, material manifestation, and cosmological dynamics no longer necessarily coincide within this framework. Field tension can exist without expressing itself in the form of compact baryonic objects. Conversely, visible matter is particularly stable where it is embedded in coherent field-mechanical structures. Large-scale expansion and the observed redshift are also not understood as the effect of an external principle, but rather as expressions of different field-mechanical regimes, especially the relaxation dynamics in structurally weak regions and the path-dependent rarefaction of radiation excitations. Many classical dark matter and dark energy problems thus lose their anomalous character and appear as different facets of a unified structural principle: gravitation

does not primarily follow mass, nor does expansion follow an extraneous substance; instead, both follow the organization, saturation, and decompression of field patterns.

This approach makes no claim to replace existing equations or to provide a complete dynamics. It introduces no new particles and does not simply explain away dark matter or dark energy. Instead, it demonstrates that the prevailing findings are ontologically ambiguous and that a field-mechanical framework can interpret the same data with fewer categorical additional assumptions. Crucially, the consequences are verifiable: UQSH implies that gravitational substructure is preferentially baryonic and geometrically influenced, that halos represent continuous stress profiles without hard object boundaries, that redshift contains path-dependent higher-order contributions, and that the visibility of satellite galaxies strongly depends on the surrounding environment, dynamics, and field geometry. These statements differ in their predictions from purely mass- or expansion-based models and are, in principle, falsifiable.

14 Reach, connectivity and open research space

The Universal Quantum Foam Hypothesis is deliberately not conceived as a competing, separate discipline, but rather as an overarching interpretive framework that unites different physical descriptions on a common ontological foundation. It neither contradicts general relativity nor quantum field theory, but instead interprets their formal structures as effective descriptions of specific states of the same medium. Spacetime curvature, field quanta, energy, momentum, and even particles appear in this view not as primary entities, but as stable organizational forms of a dynamic space field.

This approach opens up an unusual range of possibilities. Phenomena that have previously been treated as separate problem areas—dark matter, dark energy, gravity, inertia, radiation, and scale relativity—can be located within a shared conceptual space. What previously appeared as an additive inventory of the universe becomes a morphology of a single medium.

This perspective shifts the focus of future research. Instead of primarily searching for new particles, the question of the mechanical properties of space itself moves to the forefront: What elasticity does the field possess? How do stress states saturate? Under what conditions do stable knots form? How do radiation, matter, and large-scale geometry couple with each other in the long term? The UQSH does not call for reinventing established experiments, but rather for reinterpreting them. Rotation curves, lens profiles, filament geometries, background radiation, and galaxy statistics become direct measurements of an active medium.

In this sense, the UQSH does not see itself as an alternative cosmology, but as a proposal for a different way of thinking about physics: not as a science of objects in space, but as a science of the forms, states, and tensions of a space that is itself physical. It invites us to no longer view the universe as a stage, but as an actor.

15 Methodological consequences and audit trails

The UQSH does not represent a departure from empirical science, but rather a shift in what is understood as the primary object of measurement. The focus is no longer on "invisible matter," but on the measurable reactions of a medium whose state is expressed in gravity, rotation, lensing, and pattern formation. The known observables remain the same, but their

interpretation changes. What was previously considered indirect evidence of an additional substance becomes a direct indicator of field-mechanical properties.

This leads to specific methodological questions:

How quickly do stress states of the space field relax after dynamic events such as galaxy collisions, jets, or supernovae?

Are there measurable hysteresis effects in rotation curves, lens profiles, or filament geometries that indicate field-mechanical persistence?

Can transitions between "DM-strong" and "DM-weak" regimes be systematically correlated with the radiation environment, star formation rate, or geometric embedding?

Do halos exhibit continuous profiles without sharp object boundaries, as would be expected for stress states?

Do subhalos behave statistically like coupled zones of a medium, rather than like independent gravitational bodies?

The UQSH is not vague here, but implies deviations from purely mass-based models. Where CDM expects discrete halos with clear separations, the field-mechanical framework predicts transitions, superpositions, and geometric dependencies. Where particle-based DM acts independently of scale, the UQSH expects regime changes, saturation, and redistribution. Where standard models assume additive gravity, the Qu foam expects collective effects that depend on coherence and structure.

This approach proves particularly fruitful where dynamic systems are observed: colliding galaxy clusters, galactic mergers, active nuclei, and

filamentary accretion zones. In such environments, the field is forced to reorganize stresses. UQSH anticipates time-delayed, spatially extended, and nonlocal responses of the medium. Gravitational effects should not switch instantaneously, but rather persist, shift, and gradually relax.

This transforms dark matter from an "invisible substance" that is either found or not, but rather a question about the mechanics of space itself. The research path shifts from the search for new particles to the precise measurement of geometry, dynamics, and correlations. Every high-resolution rotation curve, every lens profile, every mapping of the cosmic web thus becomes an experiment on the properties of the medium.

In this sense, the UQSH is not speculative in the classical sense. It is conservative in its data, but radical in its ontology. It does not change what we measure, but rather what we believe we are measuring. And therein lies its scientific ambition: not to invent new phenomena, but to order existing ones within a more coherent framework.

16 Scale comparison as an epistemological lever

Another key advantage of UQSH lies in the possibility of systematic scale comparisons. If space is not merely a stage, but an active, tension-bearing medium, then the classical divisions between micro-, meso-, and macro-scales lose their ontological character. The same field-mechanical principles that structure galactic halos, filaments, and voids also operate in atomic, molecular, and chemical regimes, only with different temporal rhythms and spatial resolution.

Macroscopic structures thus become "magnified experiments" of the same medium. What is only indirectly accessible at the quantum or chemi-

cal scale can become visible at cosmic scales: superposition, saturation, stabilization, relaxation, and boundary formation. Galactic filaments correspond, in field mechanics, to the same stress pathways along which charge, bonding, and reaction dynamics organize themselves on a small scale. Halos are large-scale equivalents of bound states. Voids are relaxation spaces, analogous to energetically relieved regions in microscopic systems.

This opens up a new avenue of knowledge: not only does the small explain the large, but the large also becomes a model for the small. Cosmological observations can provide clues as to how field superposition leads to stability, under what conditions nodes form, how saturation limits operate, and how relaxation occurs. These insights can be projected back onto atomic and chemical processes, in which the same mechanisms are at work, but on a temporally condensed and spatially compressed scale.

The UQSH thus connects cosmology and microphysics not through formal analogy, but through a shared ontology. The universe becomes a unified experimental field of a single medium that self-organizes across scales. Macroscopic structures therefore become not only the subject of astrophysics, but also windows into the deep mechanics of matter, bonding, and reaction.

17 $Qu(x, t)$, Qu_{Horizon} and Qu_{Void}

To formally describe the field state, the expression $Qu(x, t)$ is introduced in the UQSH, which denotes the local stress state of the Qu foam as a function of position and time. Only with this designation does the question of whether this stress state must be bounded by fixed extrema become meaningful. The introduction of $Qu(x, t)$ thus necessitates an ontological clarification: Is it a quantity with absolute limits Qu_{max} und Qu_{min} , or

does it describe a scale-dependent regime field without universal bounds?

During the conceptualization process, it became clear that fixed, scale-independent limits contradict the internal logic of the approach. The Qu foam is not a self-contained system with global maxima and minima, but rather a hierarchically nested medium whose states must always be understood relative to the scale under consideration. Therefore, fieldmechanical regimes are introduced instead of absolute limits.

Qu_{Horizon} denotes the state of maximum local saturation of the medium, as achieved in compact objects up to and including black holes. In this regime, the Qu foam operates at its elastic limit. Further stress no longer leads to local densification, but rather to a spatial expansion of the boundary region. Qu_{Void} , on the other hand, denotes the state of minimum field stress in structurally weak regions where the medium is largely relaxed and carries only slight prestress.

Both quantities are not absolute, but relational. What appears as Qu_{Horizon} on one scale can be an intermediate state on another. Similarly, Qu_{Void} is not empty space, but a weakly strained field-mechanical state. The introduction of $Qu(x, t)$ makes this relationality explicitly visible and replaces the idea of fixed limits with the description of field-mechanical state regimes. This allows black holes, galactic halos, voids, and microscopic binding states to be captured within a unified formal framework without artificially disrupting the scale structure of the field.

The text is intended as a conceptual framework. The astrophysical phenomena mentioned are well established; their interpretation here is explicitly provided in the mode of a new ontological interpretation.

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Zitat

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